



**UNIVERSITI
GEOMATIKA
MALAYSIA**

DU059(W)

INDIVIDUAL ASSIGNMENT

PRINCIPLE OF ECONOMICS

DEMAND, SUPPLY, MARKET EQUILIBRIUM AND ELASTICITY

Name	MARUFA ALI
Student number	2024.5.bba03.0011
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Lecturer name	IZANOORDINA AMIR

SECTION 1: Demand, Supply and Market Equilibrium of Coffee

The following answers apply the basic principles of demand, supply and market equilibrium to the coffee market. Each scenario identifies the relevant determinant, explains the direction of the change, and shows the effect on quantity and the appropriate curve. The curves are schematic and illustrate the direction of the shift rather than exact market data.

Scenario 1: A 10% decrease in the price of tea

Because tea and coffee are substitute goods, a lower price of tea makes tea relatively cheaper than coffee. Some consumers will switch from coffee to tea. As a result, the quantity demanded of coffee decreases at each given price, so the entire demand curve for coffee shifts to the left, from D1 to D2. This is a change in demand rather than a movement along the existing demand curve.

- Quantity demanded of coffee: Decreases.
- Demand curve: Shifts leftward (D1 to D2).

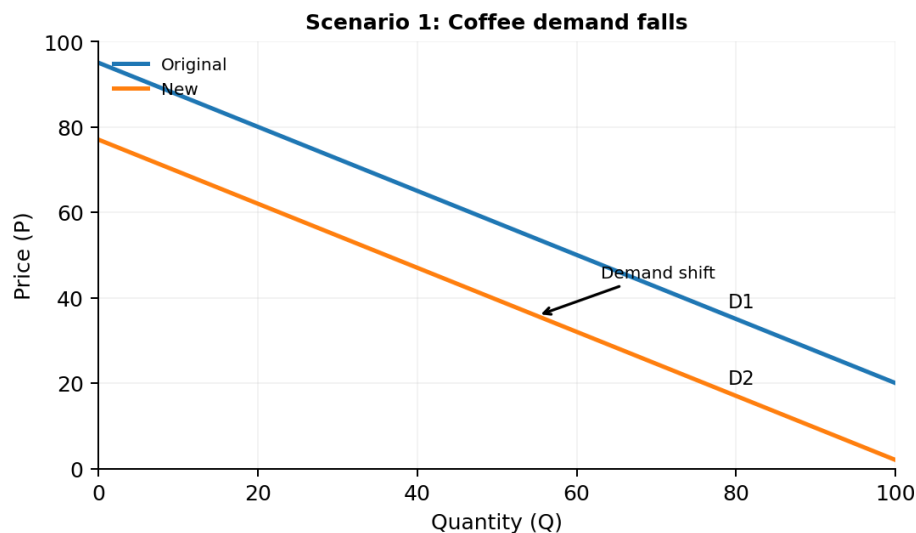


Figure 1: Schematic demand/supply curve for Scenario 1.

Scenario 2: A 5% increase in government taxes on coffee

A tax on coffee raises the effective cost of supplying the product. At each possible market price, producers are willing to supply less because part of the price is absorbed by the tax. Therefore, the supply curve shifts to the left (or upward), from S1 to S2. The quantity supplied falls at the existing market price.

- Quantity supplied of coffee: Decreases.
- Supply curve: Shifts leftward/upward (S1 to S2).

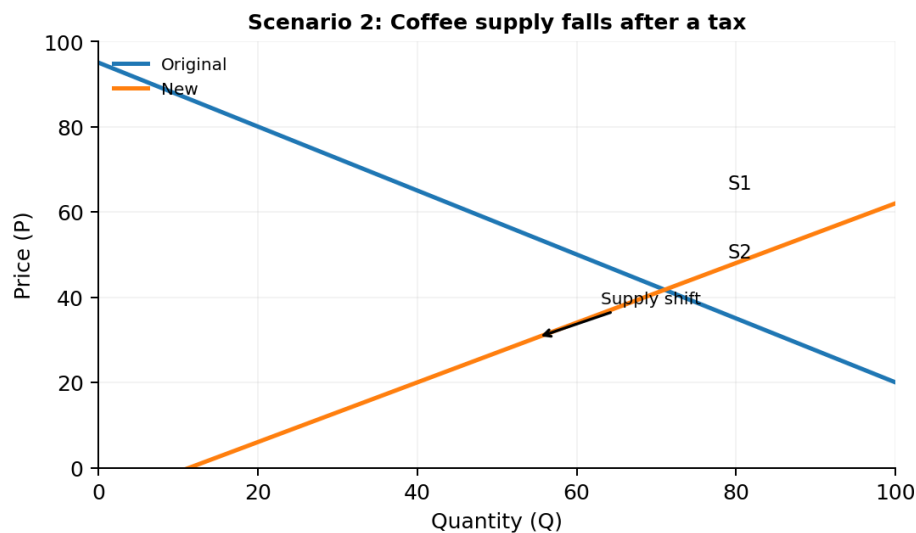


Figure 2: Schematic demand/supply curve for Scenario 2.

Scenario 3: Firms adopt advanced technology

Advanced production technology can increase productivity and reduce the cost of producing each unit of coffee. Producers can therefore supply more coffee at every price. The supply curve shifts to the right, from S1 to S2. This represents an increase in supply rather than a movement along the original supply curve.

- Quantity supplied of coffee: Increases.
- Supply curve: Shifts rightward (S1 to S2).

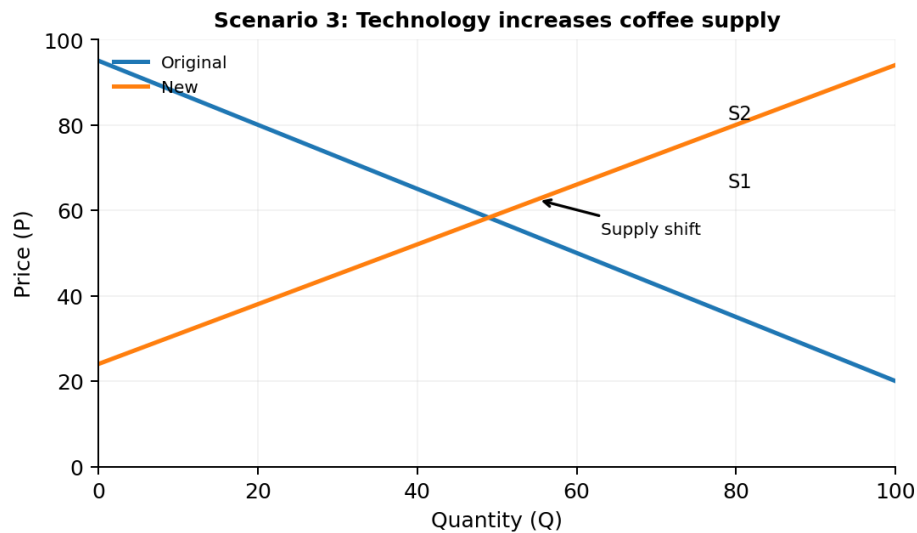


Figure 3: Schematic demand/supply curve for Scenario 3.

Scenario 4: Suppliers expect the price of coffee to decrease in three months

If suppliers expect coffee to sell for a lower price in the near future, they have an incentive to sell more coffee in the current period rather than hold inventory for later. Current supply therefore increases. The present supply curve shifts to the right, from S1 to S2.

- Quantity supplied today: Increases.
- Supply curve today: Shifts rightward (S1 to S2).

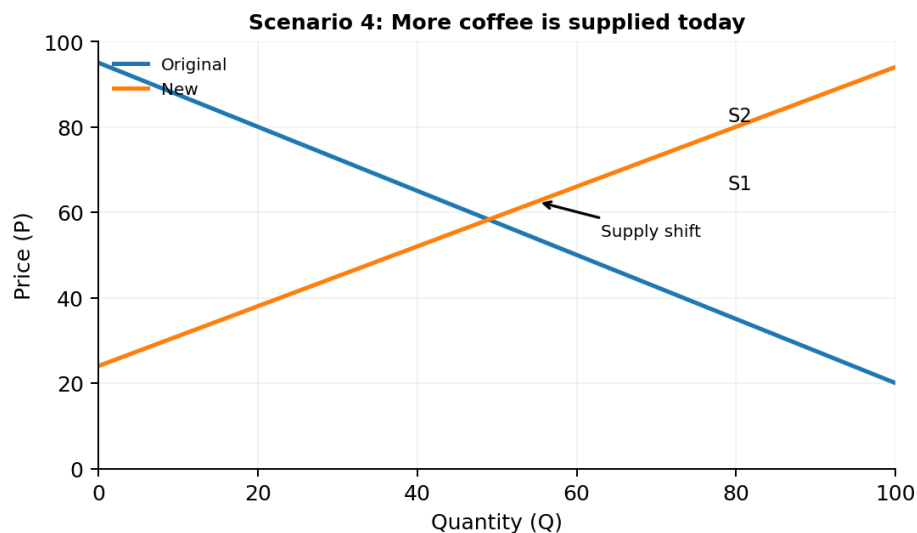


Figure 4: Schematic demand/supply curve for Scenario 4.

Scenario 5: The number of coffee buyers increases by 30% after COVID-19

An increase in the number of buyers increases market demand because more consumers are willing and able to purchase coffee. At every market price, total quantity demanded is higher. Therefore, the demand curve shifts to the right, from D1 to D2.

- Quantity demanded of coffee: Increases.
- Demand curve: Shifts rightward (D1 to D2).

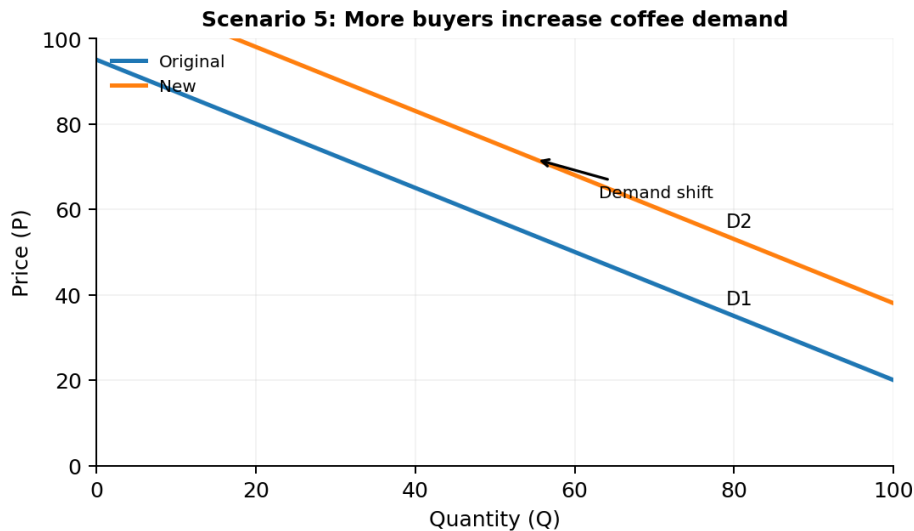


Figure 5: Schematic demand/supply curve for Scenario 5.

Scenario 6: Tea price decreases and tea and coffee are substitutes

A lower tea price makes tea more attractive relative to coffee. Demand for coffee therefore decreases, shifting the coffee demand curve to the left. With the supply curve unchanged, the new equilibrium is reached at a lower price and a lower quantity. Therefore, both equilibrium price and equilibrium quantity fall.

- Equilibrium price (P_e): Decreases.
- Equilibrium quantity (Q_e): Decreases.

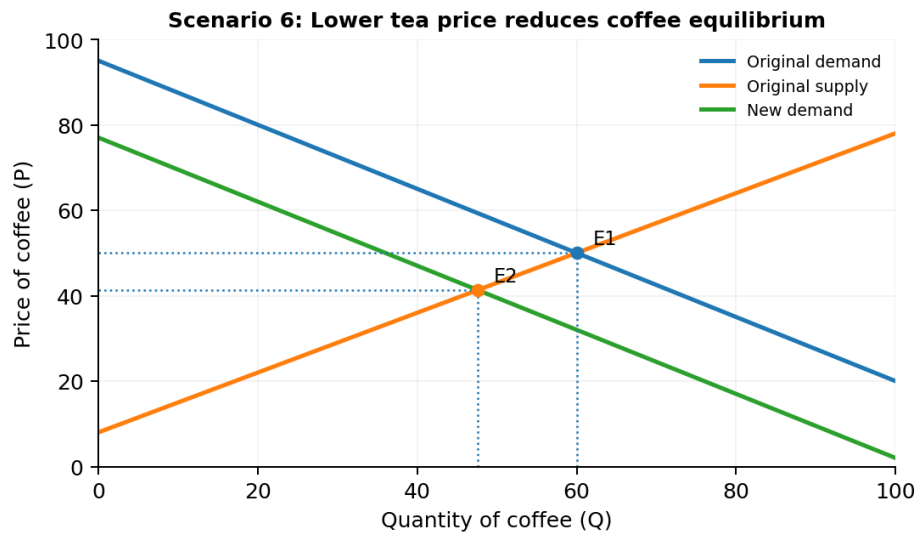


Figure 6: Schematic demand/supply curve for Scenario 6.

Scenario 7: Consumers expect the price of coffee to decrease because of a promotion

An expected lower future price can make consumers delay purchases. At present prices, some consumers postpone their coffee purchases, reducing current demand. The demand curve therefore shifts left. With supply unchanged, the new market equilibrium has a lower equilibrium price and a lower equilibrium quantity.

- Equilibrium price (P_e): Decreases.
- Equilibrium quantity (Q_e): Decreases.

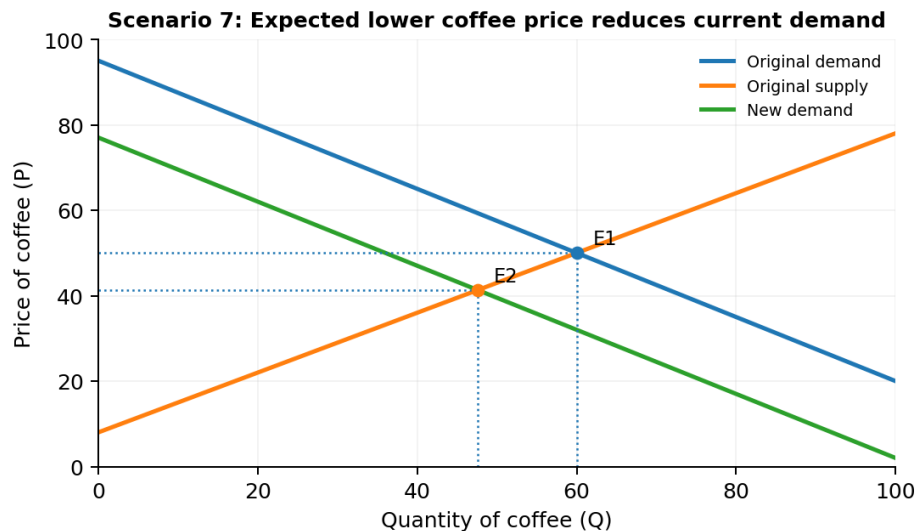


Figure 7: Schematic demand/supply curve for Scenario 7.

Scenario 8: The price of coffee beans doubles because fertilizer costs increase

Coffee beans are an important input in coffee production. When their price doubles, the cost of producing coffee rises. Producers are therefore willing to supply less at each market price, shifting the supply curve leftward. With demand unchanged, the equilibrium moves to a higher price and a lower quantity.

- Equilibrium price (P_e): Increases.
- Equilibrium quantity (Q_e): Decreases.

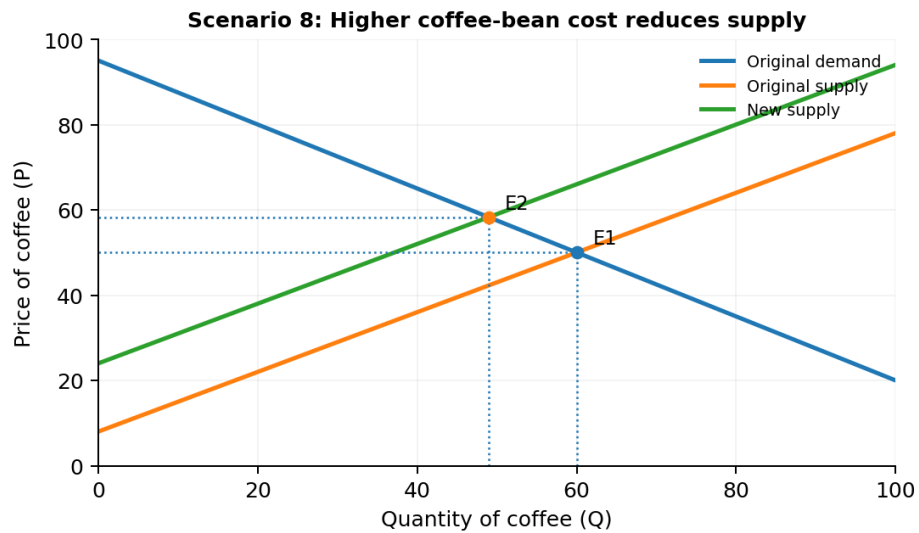


Figure 8: Schematic demand/supply curve for Scenario 8.

SECTION 2: ELASTICITY OF DEMAND AND SUPPLY

The table below is reproduced from the assignment data. The elasticity calculations use the midpoint (arc elasticity) formula where two price or income points are given: Elasticity = percentage change in quantity divided by percentage change in the relevant variable.

Price of goods X (RM)	Qd X	Qd Y	Qd Z	Quantity supply	Income (RM)
1	120	35	20	10	2000
2	90	40	20	20	1500
3	60	45	20	30	1000
4	30	50	20	40	500

a) Price elasticity of demand for goods X when price increases from RM2 to RM4

Initial price (P1) = RM2; Final price (P2) = RM4. Initial quantity (Q1) = 90; Final quantity (Q2) = 30.

Using the midpoint formula:

$$\% \text{ change in quantity demanded} = (30 - 90) / [(30 + 90)/2] = -60 / 60 = -1.00 = -100\%$$

$$\% \text{ change in price} = (4 - 2) / [(4 + 2)/2] = 2 / 3 = 0.6667 = 66.67\%$$

$$\text{Price elasticity of demand (PED)} = -1.00 / 0.6667 = -1.50$$

Therefore, the PED is -1.50, and its absolute value is 1.50.

b) Interpretation of the elasticity

Since the absolute value of PED is greater than 1 ($|\text{PED}| = 1.50$), demand for goods X is elastic over this price range. A given percentage change in price produces a proportionately larger percentage change in quantity demanded.

c) Cross-price elasticity for goods Y when the price of goods X decreases from RM4 to RM1

At $P(X) = \text{RM}4$, quantity demanded of $Y = 20$. At $P(X) = \text{RM}1$, quantity demanded of $Y = 20$.

$$\% \text{ change in quantity demanded of } Y = (20 - 20) / [(20 + 20)/2] = 0$$

$$\% \text{ change in price of } X = (1 - 4) / [(1 + 4)/2] = -3 / 2.5 = -1.20 = -120\%$$

$$\text{Cross-price elasticity of demand} = 0 / (-1.20) = 0.00$$

d) Relationship between goods Y and goods X

The cross-price elasticity is zero. Therefore, the quantity demanded of goods Y does not respond to the change in the price of goods X in the data provided. The two goods are treated as independent or unrelated goods for this dataset, rather than substitutes or complements.

e) Income elasticity of demand for goods Y when income increases from RM500 to RM1,500

When income rises from RM500 to RM1,500, quantity demanded of goods Y remains 20 at the relevant observations.

$$\% \text{ change in quantity demanded of } Y = (20 - 20) / [(20 + 20)/2] = 0$$

$$\% \text{ change in income} = (1500 - 500) / [(1500 + 500)/2] = 1000 / 1000 = 1.00 = 100\%$$

$$\text{Income elasticity of demand} = 0 / 1.00 = 0.00$$

f) Type of goods Y based on the income elasticity

The income elasticity is zero. This means that, within the observations supplied, demand for goods Y is not responsive to changes in income. The data therefore do not show goods Y as either an income-increasing normal good or an income-decreasing inferior good; its measured income elasticity is zero.

g) Price elasticity of supply for goods Y when its price decreases from RM3 to RM2

Using the quantity-supply column in the assignment table: at price RM3, quantity supplied = 45; at price RM2, quantity supplied = 40.

$$\% \text{ change in quantity supplied} = (40 - 45) / [(40 + 45)/2] = -5 / 42.5 = -0.1176 = -11.76\%$$

$$\% \text{ change in price} = (2 - 3) / [(2 + 3)/2] = -1 / 2.5 = -0.40 = -40\%$$

$$\text{Price elasticity of supply (PES)} = -0.1176 / -0.40 = 0.294 \approx 0.29$$

Therefore, the price elasticity of supply is approximately 0.29. Supply is inelastic over this price range because the absolute elasticity is less than 1.